

# Aerodynamic characterization of hollow conical shells in free fall: the influence of apex angle on subsonic drag\*

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## Abstract

The drag coefficient of a hollow conical shell falling apex-down is governed by its internal apex angle  $\alpha$ . I measured terminal velocities for five paper cones with  $\alpha = 30^\circ, 60^\circ, 90^\circ, 120^\circ$  and  $150^\circ$ , holding the mass at 5.00 g and the base radius at 6.0 cm so that geometry was the only independent variable, over  $10^4 \lesssim Re \lesssim 3 \times 10^4$ . The drag coefficient rises from  $C_d = 0.528$  at  $30^\circ$  to 1.147 at  $150^\circ$  and follows the semi-empirical form  $C_d(\alpha) = 0.6956 \sin^2(\alpha/2) + 0.4796$  with  $R^2 \approx 0.99$ . The strictly positive intercept isolates a baseline of base-pressure and viscous drag that classical Newtonian impact theory cannot produce, since that theory forces  $C_d \rightarrow 0$  as  $\alpha \rightarrow 0$ . Steady RANS simulations with the Menter  $k-\omega$  SST closure reproduce the measurements to within 0.6% at  $60^\circ$  and  $90^\circ$ , 1.8% at  $150^\circ$  and 8.1% at  $120^\circ$ , all well inside the experimental uncertainty of 17.7% to 25.4%. The  $30^\circ$  cone is the exception: the solver returns  $C_d = 0.384$  against a measured 0.528, an error of  $-27.3\%$ . I attribute this to fluid-structure interaction in the thin, flexible shell, which a steady rigid-body formulation has no mechanism to represent.

## 1 Introduction

Aerodynamic drag on a macroscopic body moving through air at moderate subsonic speed follows a quadratic velocity law,

$$F_D = \frac{1}{2} \rho C_d A v^2, \quad (1)$$

in which the drag coefficient  $C_d$  absorbs the dependence on shape and flow regime, the density  $\rho$  and the reference area  $A$  being fixed by the medium and the body [1]. Two bodies of identical frontal area can differ in  $C_d$  by more than an order of magnitude according to where the flow separates and how wide the resulting wake is; a flat plate normal to the stream reaches  $C_d \approx 1.28$  while a streamlined symmetric profile of the same frontal area reaches  $C_d \approx 0.045$  [2]. In vertical free fall the body accelerates until drag balances weight, and the resulting terminal velocity,

$$v_t = \sqrt{\frac{2mg}{\rho A C_d}}, \quad (2)$$

scales as  $C_d^{-1/2}$  at fixed mass  $m$ , density and area. A drop test that holds  $m$ ,  $\rho$  and  $A$  constant therefore measures  $C_d$  directly.

A hollow cone falling apex-down is a convenient body for isolating geometry. The sharp basal edge fixes the separation line geometrically rather than by viscous balance, so that above

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\*The experimental campaign was originally conducted in 2024 at V Liceum Ogólnokształcące, Wrocław, under the supervision of Natalia Buczak, and was awarded first place in the 20th Open Inter-School Physics Competition named after Bożena Koronkiewicz. It was extended in 2026 with uncertainty propagation, CFD validation and regression analysis.

$Re \sim 10^4$  the drag coefficient of a sharp-cornered bluff body becomes almost independent of Reynolds number [3, 4]. Unlike a solid cone, a hollow shell cannot recover pressure along a tapering afterbody; separation is forced at the base edge and the pressure drag is set by the wake behind it [5, pp. 75–89], [6, 3]. The apex angle  $\alpha$  is then left as the single geometric parameter. This is the classical falling paper cone problem, a standard benchmark for the relation between bluff-body geometry and steady descent [7].

The question addressed here is how  $\alpha$  governs  $v_t$  and  $C_d$ . I answer it by measuring terminal velocities for five cone geometries under controlled mass and frontal area, by simulating the same five geometries at the measured descent velocities, and by testing both datasets against the Newtonian impact prediction  $C_d = 4 \sin^2(\alpha/2)$  and against a semi-empirical correction to it. The measurement uncertainties are propagated in full, and the predicted scaling  $v_t \propto C_d^{-1/2}$  is checked against the data rather than assumed.

The scope is limited to thin-walled, low-rigidity hollow shells falling apex-down in incompressible subsonic flow at  $10^4 \lesssim Re \lesssim 3 \times 10^4$ . Only stable or quasi-steady descent segments are analysed, since the orientation of a falling cone can switch, tilt or oscillate depending on apex angle and Reynolds number [8, 7], and the steady force balance of Equation (2) does not apply to those states. The numerical work is restricted to steady incompressible Reynolds-Averaged Navier–Stokes (RANS) solutions with a single turbulence closure. The results do not extend to apex-up or inclined orientations, to solid cones, or to deliberately rotating bodies.

## 2 Theoretical framework

### 2.1 Force balance in free fall

A cone released from rest accelerates under its weight  $F_g = mg$  until the drag of Equation (1) grows to match it. Buoyancy is neglected: with  $\rho_{\text{air}}/\rho_{\text{cone}} \ll 1$  the buoyant force is three orders of magnitude below the weight. Taking downwards as positive, the steady-state condition is

$$F_g - F_D = 0 \quad \Rightarrow \quad mg = \frac{1}{2} \rho C_d A v_t^2, \quad (3)$$

whose solution is Equation (2). Because  $m$ ,  $\rho$  and  $A$  are held fixed across the specimen set, any measured difference in  $v_t$  maps directly onto a difference in  $C_d$  produced by geometry alone [1, 9].

### 2.2 Newtonian impact theory

The classical baseline for the dependence of  $C_d$  on cone geometry is Newton’s corpuscular model, which treats the fluid as a stream of non-interacting particles moving in straight lines at constant speed  $v$  relative to a stationary cone [10, 7]. It assumes that collisions with the surface are perfectly elastic, that particles do not collide with one another, that the flow carries no viscosity, boundary layer, turbulence or wake, and that drag arises solely from the momentum balance along the flow direction.

Consider a surface element  $dA$  struck by particles of mass  $m_p$  at speed  $v$ , and let  $\gamma$  be the local angle between the stream direction and the surface normal, as defined in Figure 1. In an elastic collision the normal velocity component reverses, so the momentum change of a single particle along the flow axis is proportional to  $\cos^2 \gamma$ , while the transverse components cancel by rotational symmetry. The number of particles reaching  $dA$  per unit time is proportional to  $v \cos \gamma$ . Combining the two contributions and using  $\rho = N m_p$  for a particle number density  $N$  gives

$$dF = 2\rho v^2 \cos^3 \gamma dA, \quad F_D = 2\rho v^2 \iint_S \cos^3 \gamma dA. \quad (4)$$

The geometric factor  $I_{\text{geo}} = \iint_S \cos^3 \gamma \, dA$  follows from the cone's linear slant profile  $y(x) = ax$ , for which  $y'(x) = a$  is constant [10]:

$$\frac{I_{\text{geo}}}{\pi} = \int_0^r \frac{2x}{1 + y'(x)^2} dx = \int_0^r \frac{2x}{1 + a^2} dx = \frac{r^2}{1 + a^2}. \quad (5)$$

With  $A = \pi r^2$  and  $1/(1 + a^2) = 1/(1 + \tan^2 \gamma) = \cos^2 \gamma$ , and with the wall inclination related to the internal apex angle by  $\gamma = \pi/2 - \alpha/2$ ,

$$\cos^2 \gamma = \cos^2 \left( \frac{\pi}{2} - \frac{\alpha}{2} \right) = \sin^2 \left( \frac{\alpha}{2} \right) \implies I_{\text{geo}} = A \sin^2 \left( \frac{\alpha}{2} \right). \quad (6)$$

Substituting back into Equation (4) and comparing with Equation (1) yields

$$F_D = 2\rho v^2 A \sin^2 \left( \frac{\alpha}{2} \right), \quad C_d = 4 \sin^2 \left( \frac{\alpha}{2} \right). \quad (7)$$

The standard hypersonic form of Newtonian theory assumes inelastic collisions, in which the normal momentum component is absorbed rather than reversed, and returns a leading coefficient of 2 instead of 4. The functional dependence on  $\sin^2(\alpha/2)$  is identical in both, and so are the limitations examined next [5, pp. 877–881].

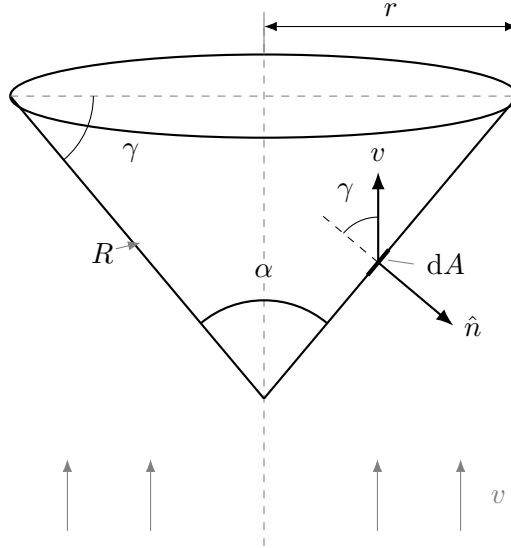


Figure 1: Cone geometry and the angles used in the Newtonian derivation, with internal apex angle  $\alpha$ , base radius  $r$ , slant height  $R$ , and the angle  $\gamma = \pi/2 - \alpha/2$  between the stream direction and the surface normal  $\hat{n}$ .

### 2.3 Limitations of the Newtonian model and a semi-empirical correction

Because  $\sin^2(\alpha/2) \in [0, 1]$ , Equation (7) confines  $C_d$  to  $[0, 4]$  and drives it to zero as  $\alpha \rightarrow 0$ . Real slender bodies retain a finite drag from surface friction alone, so the zero-drag limit is unphysical [5, pp. 75–89]. The model also treats the region behind the body as a passive shadow: it assigns no pressure to the base, and therefore carries no base-pressure deficit and no recirculation, both of which dominate subsonic bluff-body drag [3]. It further ignores skin friction and the upstream stagnation field in which streamlines deflect before reaching the surface [5, pp. 75–89].

Retaining the geometric scaling of the collision model while admitting the physics it omits gives a two-parameter semi-empirical form [5, pp. 75–89], [7]:

$$C_d = a \sin^2 \left( \frac{\alpha}{2} \right) + b. \quad (8)$$

The slope  $a$  measures how fast pressure drag and separation width grow with apex angle; the intercept  $b$  is the baseline of skin friction and base pressure that survives as  $\alpha \rightarrow 0$  and removes the non-physical zero-drag prediction [7].

### 3 Methodology and experimental setup

#### 3.1 Test specimens

Five thin-walled hollow conical shells were folded from paper with internal apex angles  $\alpha \in \{30^\circ, 60^\circ, 90^\circ, 120^\circ, 150^\circ\}$  (Figure 2). All five share a base radius  $r = 6.0(1)$  cm, which fixes the frontal projected area at  $A = 0.011\,31(38)$  m<sup>2</sup> for the apex-down configuration, and all five were ballasted to  $m = 5.00(1)$  g with a small mass of plasticine placed inside the apex. The ballast also lowered the centre of mass and stabilised the descent orientation; the mass of the joining tape was compensated by plasticine on the opposite side to limit centre-of-mass eccentricity. Fixing  $m$  and  $A$  leaves  $\alpha$  as the only independent variable, so that differences in  $v_t$  can be attributed to  $C_d$  through Equation (3) [1, 7].

Four cones ( $60^\circ$  to  $150^\circ$ ) were cut from 200 g/m<sup>2</sup> poster board of thickness 0.3 mm. The  $30^\circ$  geometry has by far the largest lateral-to-base area ratio of the set and buckled macroscopically in that stock, so it was made instead from 125 g/m<sup>2</sup> paper of thickness 0.2 mm. This is the one controlled parameter that could not be held constant across the set, and Section 6.2 returns to its consequences.

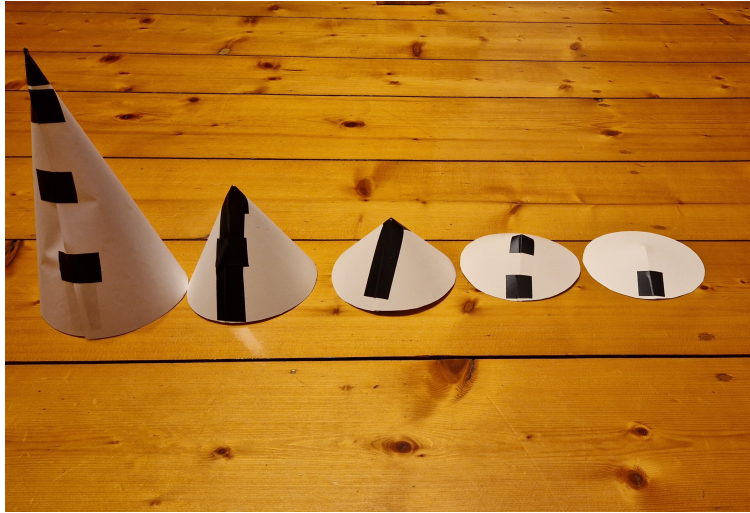


Figure 2: The five hollow cones used in the study, from left  $30^\circ$ ,  $60^\circ$ ,  $90^\circ$ ,  $120^\circ$  and  $150^\circ$ .

#### 3.2 Test stand and drop geometry

Drops were recorded against a vertical uniform wall carrying three horizontal high-contrast fiducial markers spaced at 0.50 m with a placement uncertainty of  $\delta y = \pm 0.01$  m (Figure 3). The three markers define a 1.0 m measurement window split into two independent 0.50 m intervals, A–B and B–C, which allows the constancy of the descent speed to be tested within a single trial.

Trajectories were recorded with a Nikon D850 and a Nikon AF-S NIKKOR 70–200 mm f/2.8G ED VR II lens placed 4.5 m from the wall. The long focal length and the stand-off distance together approximate an orthographic projection, which limits perspective distortion and parallax at the edges of the window. The optical axis was set perpendicular to the wall and aimed at the centre marker so that the image scale is the same at the top and bottom of the window. Recording ran at 60 fps, giving a frame interval of  $\delta t \approx 16.7$  ms.

Specimens were released from rest apex-down at approximately 3.3 m above the lowest marker, so that they covered an acceleration distance of about 2.3 m before entering the window. For quadratic drag the approach to terminal velocity over a fallen distance  $y$  is

$$\frac{v(y)}{v_t} = \sqrt{1 - \exp\left(-\frac{2gy}{v_t^2}\right)}. \quad (9)$$

At  $y = 2.3$  m and the measured terminal velocities, Equation (9) gives 98.1%  $v_t$  for the 30° cone, 99.1% at 60°, 99.8% at 90° and above 99.9% at 120° and 150°, comfortably above the 95% threshold adopted as the criterion for steady descent in the window. The same distance gives aerodynamic restoring moments time to damp the angular perturbation of a manual release.

The local air density was  $\rho = 1.1937$  kg/m<sup>3</sup>, obtained from the ideal gas law using the ambient pressure, temperature and relative humidity recorded alongside the drop tests [11].

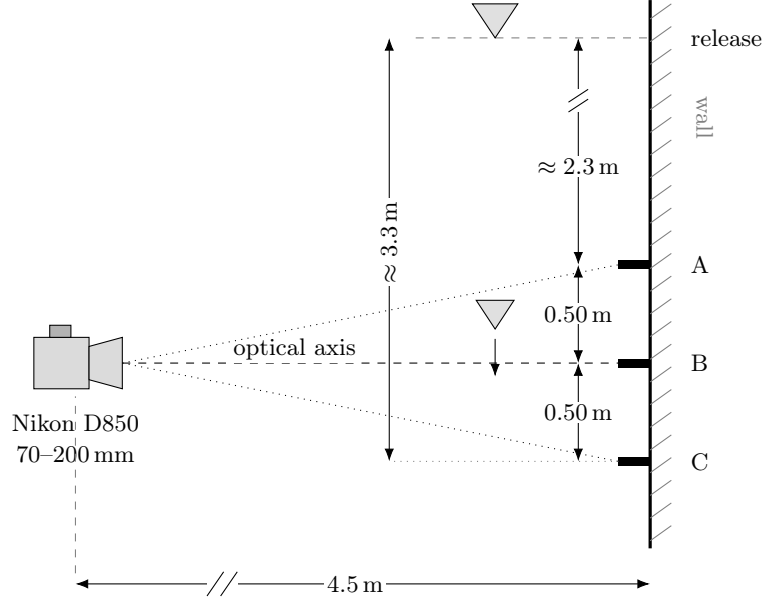


Figure 3: Test stand geometry: camera stand-off from the calibration wall, the three fiducial markers bounding the two 0.50 m measurement intervals, and the release height above the window. Horizontal and vertical scales differ and the broken spans are shortened.

### 3.3 Measurement protocol

Ten drops were recorded per geometry, and the first six showing purely vertical quasi-steady descent were accepted. Trials showing vortex-induced flutter, tumbling or lateral drift were rejected, because in those states the force balance of Equation (3) does not hold and no single  $v_t$  can be assigned [7, 8]. The complete log, including rejected and censored trials, is given in Appendix A.

Footage was analysed frame by frame in Adobe Premiere Elements 2023, tracking the cone apex against the scaled background. For each 0.50 m interval the frame count  $n$  gives the transit time  $t = n/60$  and the interval speed  $v = y/t$ . Attainment of terminal velocity was tested within each trial by requiring  $t_{AB}$  and  $t_{BC}$  to agree to within the frame discretisation error,  $\delta t \leq 1/60$  s, that is within one frame. No accepted trial exceeded that threshold. For each geometry the six accepted speeds were averaged to give  $v_t$ .

### 3.4 Uncertainty propagation

Three quantities are measured directly and carry instrumental uncertainty: the marker spacing,  $\delta y = \pm 0.01$  m; the base radius,  $\delta r = \pm 0.1$  cm; and the transit time, limited by the frame interval to  $\delta t = \pm 1/60$  s. Uncertainties in the derived quantities were combined by the root-sum-square rule for independent sources [12, 13].

For the speed on a calibration interval,  $v = y/t$ , the quotient rule gives

$$\frac{\delta v}{v} = \sqrt{\left(\frac{\delta y}{y}\right)^2 + \left(\frac{\delta t}{t}\right)^2}. \quad (10)$$

The time term dominates, and it dominates most for the fastest specimen because  $\delta t$  is fixed while  $t$  shrinks:  $\delta v/v$  runs from 12.5% at 30° to 8.6% at 150°. The area follows the power rule,  $\delta A/A = 2\delta r/r \approx 3.3\%$ , constant across the set. The mass contributes  $\delta m/m = 0.0001/0.005 = 2.0\%$ . Ambient conditions were monitored but not logged continuously, so a conservative  $\delta\rho/\rho = 1\%$  was assumed.

Rearranging Equation (3) gives  $C_d = 2mg/(\rho Av^2)$ , and the product rule for relative uncertainties gives

$$\frac{\delta C_d}{C_d} = \sqrt{\left(\frac{\delta m}{m}\right)^2 + \left(\frac{\delta A}{A}\right)^2 + \left(\frac{\delta \rho}{\rho}\right)^2 + \left(2\frac{\delta v}{v}\right)^2}. \quad (11)$$

Velocity enters the drag force quadratically and therefore enters Equation (11) with a factor of two, which makes it the dominant term by a wide margin. The resulting totals are collected in Table 1 and range from 17.7% at 150° to 25.4% at 30°.

Table 1: Relative propagated uncertainties used throughout the analysis.

| $\alpha$ [°] | $\delta v/v$ [%] | $\delta A/A$ [%] | $\delta m/m$ [%] | $\delta \rho/\rho$ [%] | $\delta C_d/C_d$ [%] |
|--------------|------------------|------------------|------------------|------------------------|----------------------|
| 30           | 12.5             | 3.3              | 2.0              | 1.0                    | 25.4                 |
| 60           | 11.4             | 3.3              | 2.0              | 1.0                    | 23.1                 |
| 90           | 9.9              | 3.3              | 2.0              | 1.0                    | 20.2                 |
| 120          | 9.4              | 3.3              | 2.0              | 1.0                    | 19.2                 |
| 150          | 8.6              | 3.3              | 2.0              | 1.0                    | 17.7                 |

The apex angle carries its own construction uncertainty. Cone layout is governed by the relation  $\sin(\alpha/2) = r/R$ , with both the base radius  $r$  and the slant height  $R$  held to  $\pm 0.1$  cm. Propagating those tolerances through that relation gives an error that grows sharply as the cone flattens, because a fixed deviation in  $R$  maps onto an ever larger deviation in  $\alpha$  as  $\alpha \rightarrow 180^\circ$ : from  $\delta\alpha = 0.529^\circ$  at 30° to  $9.910^\circ$  at 150°. These values appear as the horizontal error bars in all subsequent plots.

Two distinct uncertainties coexist. The instrumental uncertainty above is dominated by camera time discretisation. The statistical uncertainty of the mean over  $N = 6$  accepted trials,  $SE = \sigma/\sqrt{N}$  [14], was much smaller, at  $SE_v \approx \pm 0.04$  m/s. All error bars reported here use the larger instrumental figure, which avoids overstating the precision of the method [12, 13].

## 4 Experimental results

Of the 50 drops recorded, 5 were rejected on stability grounds: one each at 30°, 90° and 120°, and two at 150°, all for transverse trajectory divergence or kinematic instability. The 60° geometry produced no rejected trial. The remaining 15 drops were censored once the six-trial quota was met, so every geometry contributes an equal  $N = 6$  to the mean. The full log is in Appendix A.

Terminal velocity decreases monotonically with apex angle across the whole set (Table 2). The most slender cone settles at  $v_t \approx 3.71$  m/s and the bluffest at  $v_t \approx 2.52$  m/s, a reduction of 32% in descent rate. Since  $m$ ,  $\rho$  and  $A$  were fixed, that entire change is carried by  $C_d$ . The frame counts show how close the fastest specimen sits to the resolution floor: every accepted 30° trial crossed a 0.50 m interval in either 8 or 9 frames, so one frame separates the two distinct speeds the method can return there, which is the origin of the 12.5% velocity uncertainty in Table 1. The slowest specimen spans 11 to 13 frames over the same interval and is correspondingly better resolved.

Table 2: Mean terminal velocities with propagated uncertainties, rounded to three decimal places.

| $\alpha$ [°] | $\delta\alpha$ [°] | $v_t$ [m/s] | $\delta v/v$ [%] | $\delta v$ [m/s] |
|--------------|--------------------|-------------|------------------|------------------|
| 30           | 0.529              | 3.711       | 12.5             | 0.466            |
| 60           | 1.233              | 3.364       | 11.4             | 0.384            |
| 90           | 2.339              | 2.903       | 9.9              | 0.288            |
| 120          | 4.376              | 2.748       | 9.4              | 0.258            |
| 150          | 9.910              | 2.517       | 8.6              | 0.218            |

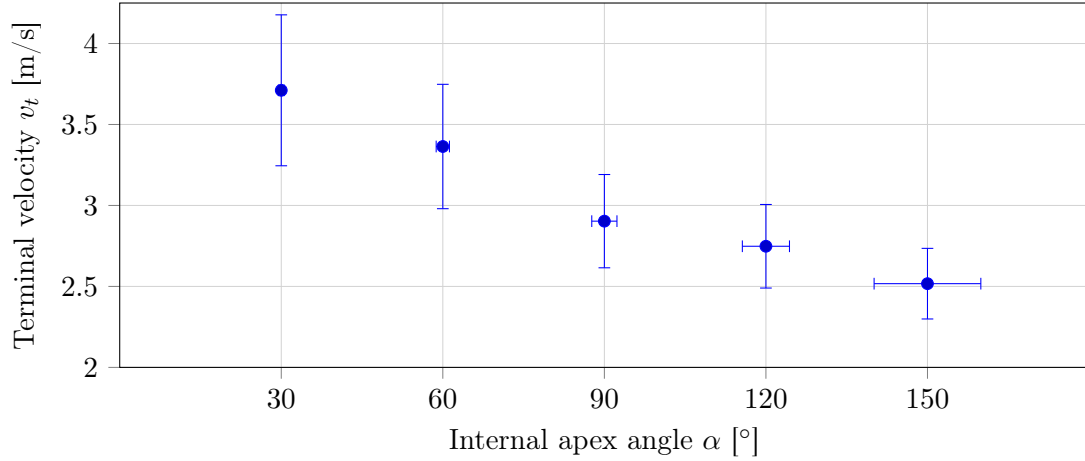


Figure 4: Measured terminal velocity against apex angle, with instrumental uncertainties in both coordinates.

Inverting Equation (3) with the constant  $m$ ,  $\rho$  and  $A$  of Section 3.1 and the mean velocities of Table 2 gives the drag coefficients in Table 3, with uncertainties from Equation (11). The coefficient more than doubles over the range tested, from  $C_d \approx 0.528$  at  $30^\circ$  to  $C_d \approx 1.147$  at  $150^\circ$ , under strictly controlled mass and frontal area.

Table 3: Experimental drag coefficients with propagated uncertainties, rounded to three decimal places.

| $\alpha$ [°] | $\delta\alpha$ [°] | $C_d$ [-] | $\delta C_d/C_d$ [%] | $\delta C_d$ [-] |
|--------------|--------------------|-----------|----------------------|------------------|
| 30           | 0.529              | 0.528     | 25.4                 | 0.134            |
| 60           | 1.233              | 0.642     | 23.1                 | 0.148            |
| 90           | 2.339              | 0.862     | 20.2                 | 0.174            |
| 120          | 4.376              | 0.962     | 19.2                 | 0.184            |
| 150          | 9.910              | 1.147     | 17.7                 | 0.203            |

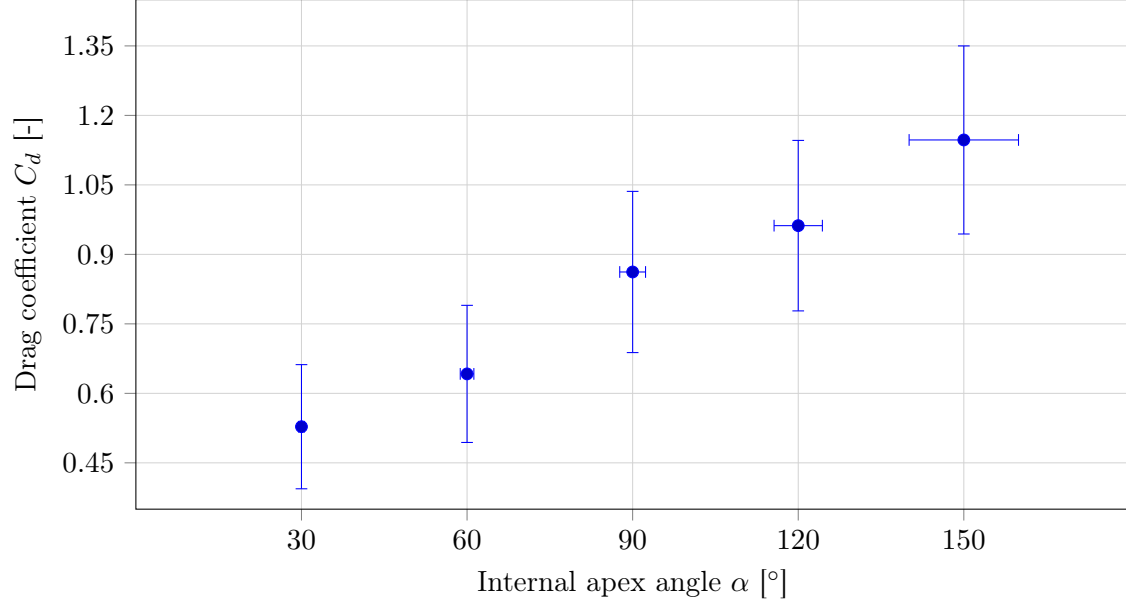


Figure 5: Experimental drag coefficient against apex angle, with error bars  $\delta\alpha$  and  $\delta C_d$  from Section 3.4.

## 5 Numerical method and results

### 5.1 Domain, boundary conditions and solver

The five geometries were rebuilt numerically with the same base radius  $r = 6.0$  cm and frontal area  $A$  as the physical specimens. The steady incompressible RANS equations were solved with SimScale, a cloud simulation platform that runs OpenFOAM-based finite-volume solvers, using the SIMPLE algorithm [15, 16, 17]. A steady formulation is appropriate because the quantity of interest is the time-averaged drag at the equilibrium  $F_D = F_g$  rather than any instantaneous force.

The boundary conditions reproduce free fall in open air rather than a wind tunnel:

- *Velocity inlet.* A uniform inlet velocity fixed at the measured  $v_t$  for the corresponding  $\alpha$ , with pressure extrapolated from the interior, which is the well-posed pairing for incompressible inflow.
- *Pressure outlet.* Fixed kinematic gauge pressure  $p/\rho = 0$  with zero normal velocity gradient, so that wake structures leave the domain without artificial reflection.
- *Far field.* Slip (symmetry) conditions on the outer radial boundaries, enforcing  $u_n = 0$  and zero normal scalar gradients. This removes both the blockage effect and the spurious wall layers that a no-slip tunnel wall would introduce.
- *Cone surface.* No-slip, resolving the near-wall velocity gradient and the skin-friction contribution to  $C_d$ .

Setting the inlet velocity to the measured  $v_t$  keeps the Reynolds number and the dynamic reference pressure identical between experiment and simulation.

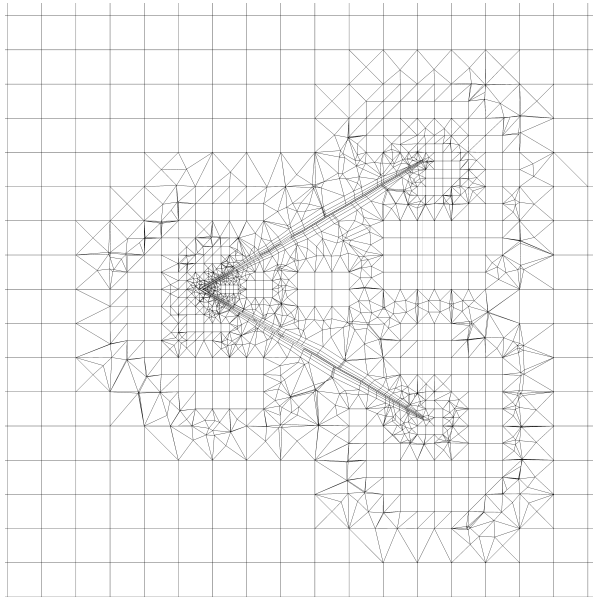
### 5.2 Turbulence closure

Menter’s  $k\text{-}\omega$  SST model was used as the closure in every case. It combines two closure families through a switching function: near the wall it takes the  $k\text{-}\omega$  form, which maps the boundary layer

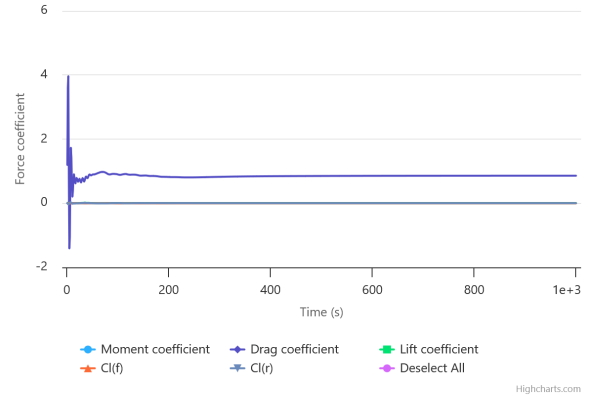
well, while in the free-stream region it switches to behaviour similar to  $k-\epsilon$ , reducing sensitivity to far-field conditions [18, 19]. That combination suits the present problem. Over the blunt part of the range, from  $60^\circ$  to  $150^\circ$ , the drag is set by separation at the base edge and by the wake behind it under an adverse pressure gradient, at  $Re \approx 10^4$  to  $3 \times 10^4$ , and a single closure applied to all five apex angles keeps the comparison between them internally consistent.

### 5.3 Mesh and convergence

Each domain was meshed with wake refinement downstream of the base and inflation layers on the cone surface (Figure 6a). Convergence was judged from the iterative history of the force coefficients together with the domain residuals, and a solution was accepted once the coefficients reached an asymptote with fluctuations below 0.1% (Figure 6b). All results below were extracted after 1000 iterations, at which point every case met that criterion.



(a) Mesh cut-plane,  $\alpha = 60^\circ$ .



(b) Convergence history,  $\alpha = 90^\circ$ .

Figure 6: Longitudinal mesh cut-plane showing wake refinement and near-wall inflation layers, and the iterative history of the force coefficients against iteration number.

### 5.4 Computed drag coefficients

The computed coefficients are listed in Table 4 and plotted in Figure 7. They rise non-linearly with apex angle, from  $C_d = 0.384$  at  $30^\circ$  to 1.167 at  $150^\circ$ , which is the trend expected as the pressure contribution from separation and wake growth takes over. Four of the five values sit close to the measurements of Table 3. The  $30^\circ$  case does not, and it is the subject of Section 6.2.

Table 4: Drag coefficients computed with the  $k-\omega$  SST closure after 1000 RANS iterations, rounded to three decimal places.

| $\alpha$ [ $^\circ$ ] | $C_{d,\text{CFD}}$ [-] |
|-----------------------|------------------------|
| 30                    | 0.384                  |
| 60                    | 0.642                  |
| 90                    | 0.857                  |
| 120                   | 1.040                  |
| 150                   | 1.167                  |

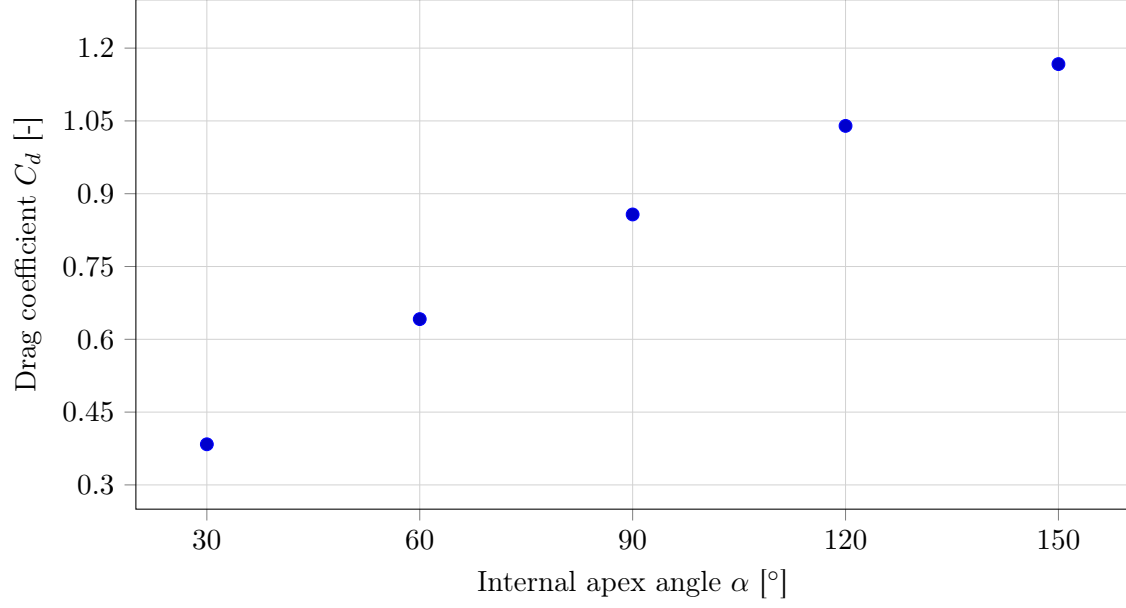


Figure 7: Computed drag coefficient against apex angle for the  $k-\omega$  SST closure.

### 5.5 Flow fields

Longitudinal cut-planes of velocity magnitude map the near-wake recirculation (Figure 8). The 150° shell leaves a wide, persistent velocity deficit; the 60° cone leaves a narrow shear layer. The corresponding pressure fields (Figure 9) show the stagnation zone at the apex, where the stream decelerates to near zero and static pressure reaches a local maximum, together with the reduced-pressure region behind the base.

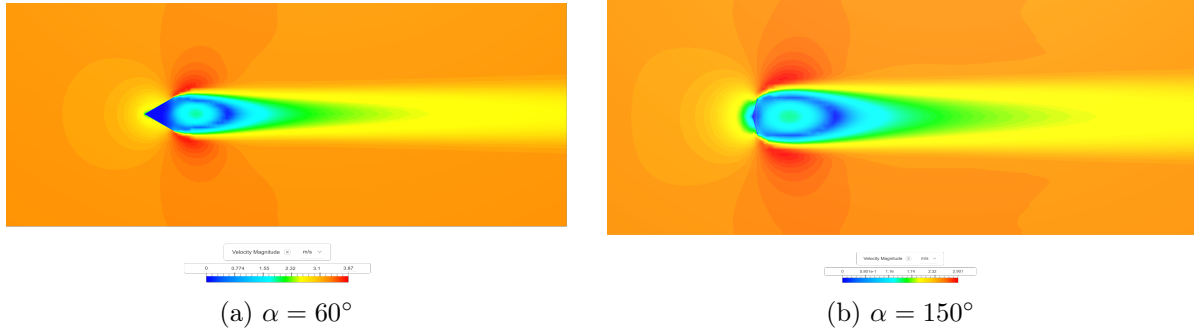


Figure 8: Velocity magnitude contours on a longitudinal section for the 60° and 150° cones.

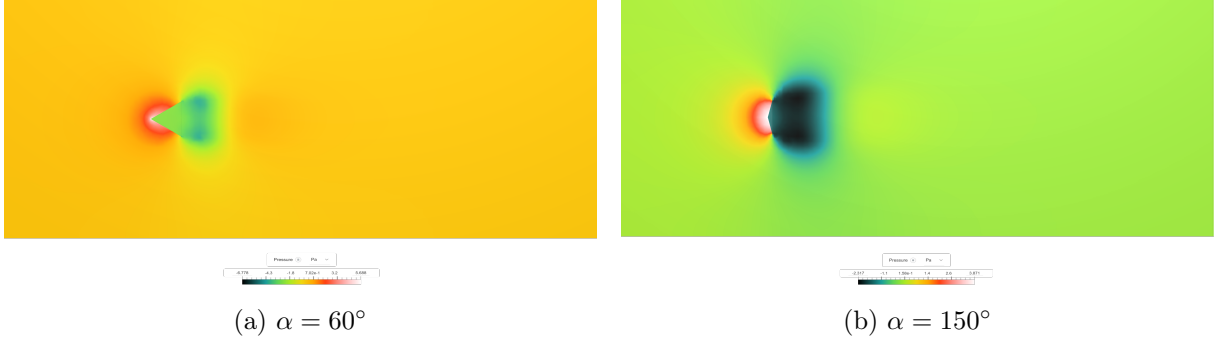


Figure 9: Pressure distribution for the  $60^\circ$  and  $150^\circ$  cones, showing the apex stagnation zone and the reduced-pressure region behind the base.

## 6 Discussion

### 6.1 Measured against computed drag

Table 5 places the two datasets side by side with the relative prediction error

$$E_{\text{rel}} = \frac{C_{d,\text{CFD}} - C_{d,\text{exp}}}{C_{d,\text{exp}}},$$

and Figure 10 plots them together. For every geometry with  $\alpha \geq 60^\circ$  the computed value falls inside the experimental error bar.

Table 5: Experimental and computed drag coefficients with the relative prediction error of the  $k\text{-}\omega$  SST solution.

| $\alpha$ [°] | $\delta\alpha$ [°] | $C_{d,\text{exp}}$ [-] | $\delta C_d/C_d$ [%] | $\delta C_d$ [-] | $C_{d,\text{CFD}}$ [-] | Relative error [%] |
|--------------|--------------------|------------------------|----------------------|------------------|------------------------|--------------------|
| 30           | 0.529              | 0.528                  | 25.4                 | 0.134            | 0.384                  | -27.3              |
| 60           | 1.233              | 0.642                  | 23.1                 | 0.148            | 0.642                  | -0.1               |
| 90           | 2.339              | 0.862                  | 20.2                 | 0.174            | 0.857                  | -0.6               |
| 120          | 4.376              | 0.962                  | 19.2                 | 0.184            | 1.040                  | +8.1               |
| 150          | 9.910              | 1.147                  | 17.7                 | 0.203            | 1.167                  | +1.8               |

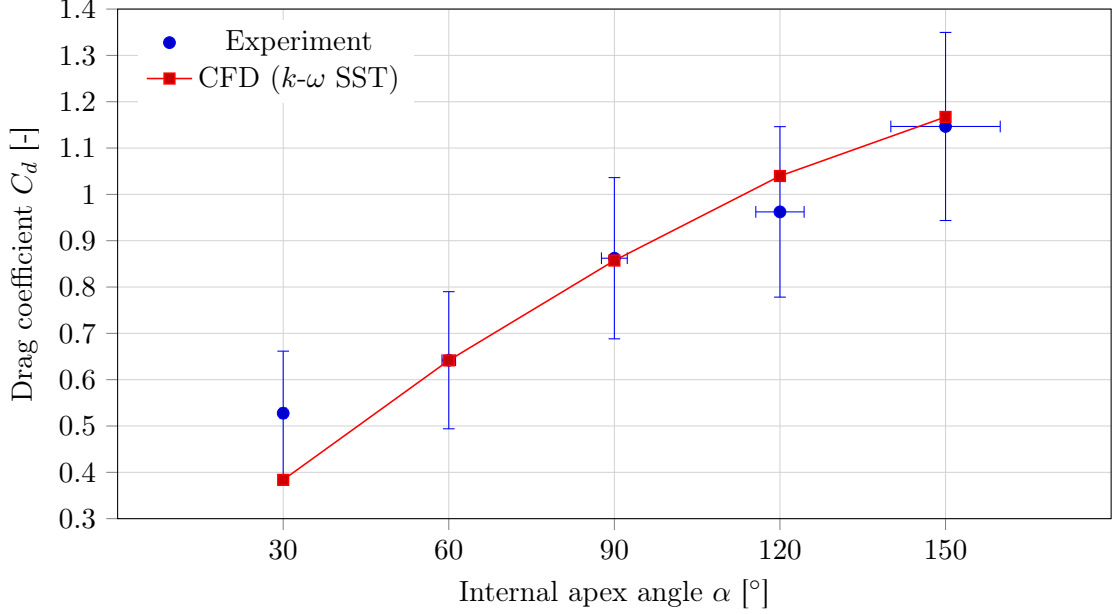


Figure 10: Measured and computed drag coefficient against apex angle, with experimental error bars.

At 60° and 90° the two methods differ by less than 1%, and at 150° by 1.8%. Those margins sit an order of magnitude below the experimental uncertainty of roughly 20%, which is dominated by the squared velocity term of Equation (11). The 120° case deviates by 8.1%, still inside its 19.2% error bar. As the cone flattens,  $C_d$  becomes more responsive to the detail of the base pressure distribution, so a moderate shift in the predicted base pressure moves  $C_d$  noticeably [12]. The 150° deviation is positive, consistent with a steady solution over-predicting wake stability for a strongly separated flow and so over-predicting pressure drag slightly relative to the average of a real descent [19, 18].

Two things explain the agreement over the blunt range. First, separation is geometrically determined. On a body with a sharp streamwise edge the separation line is fixed by the edge rather than by a viscous balance, so the drag coefficient stabilises above  $Re \gtrsim 10^4$  and can be predicted with one closure and one mesh strategy across the range [3]. Second, the closure matches the physics: the SST model was built for separated flows under adverse pressure gradients while retaining good wall behaviour from its  $k-\omega$  part and lower sensitivity to far-field conditions from its  $k-\epsilon$  part [19, 18], and pressure drag from separation is exactly what sets  $C_d$  for these geometries. Because the inlet velocity was matched to the measured  $v_t$  and the outer boundaries are slip (Section 5.1), the residual differences are attributable to the limits of averaged modelling rather than to artefacts of the domain.

## 6.2 Divergence at the slender geometry

The 30° cone is the one case where the two methods disagree beyond the measurement uncertainty. The measured coefficient is  $C_{d,\text{exp}} \approx 0.528$  against a computed  $C_{d,\text{CFD}} \approx 0.384$ , an error of −27.3%.

The discrepancy is not numerical. The 30° case used the same domain, boundary conditions and closure as the other four, and it met the same convergence criterion, with force-coefficient fluctuations below 0.1% over the final iterations (Section 5.3). What sets it apart is how well the rigid-body idealisation describes the physical specimen.

A steady RANS solution assumes a geometry fixed in time, perfectly formed and hydraulically smooth, and derives drag from averaged pressure and shear over a stationary surface. The 30°

specimen departs from those assumptions further than any other member of the set, and it does so by construction:

- It has the largest lateral-to-base area ratio of the five, and it is the only one built from 125 g/m<sup>2</sup>, 0.2 mm stock, adopted because the 0.3 mm board buckled at that aspect ratio (Section 3.1). It is the least rigid sample tested, and its flexibility follows from its slenderness rather than from an incidental fabrication defect.
- A shell that flexible undergoes micro-vibration in free fall. Small out-of-plane motion of the wall modulates the local surface incidence, disturbs the boundary layer, and makes the instantaneous body geometry unsteady. Thin bodies falling through a viscous medium are known to enter fluttering descent modes through this coupling between the vortex wake and the motion of the body [20].
- The time-averaged consequence is a wider wake. The nominal reference area is fixed by the base radius, but an oscillating surface presents a greater average bluntness to the flow. Pressure drag on a bluff body is governed by the extent and depth of the low-pressure region behind the base, so a modest widening of that region raises  $C_d$  appreciably [3].
- Lower-grammage paper is also more prone to local creasing, which thickens the near-wall layer and shifts the separation conditions away from those of a smooth surface.

None of these mechanisms exists in the numerical model. A steady solver cannot represent a surface that moves, and a rigid geometry cannot generate the extra wake width that motion produces. The computed value is therefore expected to fall short: the solver returns the drag of a perfectly rigid 30° cone, while the experiment returns the drag of a flexible one.

Two features of the dataset bound this reading rather than merely assume it. The divergence appears only where the idealisation is weakest: the blunt cones are stiffer, have far smaller lateral-to-base area ratios, and separate at a geometrically fixed edge, and all four agree with the computation inside the measurement uncertainty. The disagreement is also one-sided, with the experiment above the computation, which is the direction that fluid-structure coupling predicts; a random experimental error would carry no such sign. The interpretation is testable, and Section 7 states the test.

### 6.3 Wake topology and the failure of Newtonian theory

For the blunt geometries the resistance is dominated by form drag arising from the pressure difference between the windward face and the leeward base. At 150° the flow cannot accelerate around the steep flanks, so near-stagnation pressure extends across the whole projected frontal area instead of concentrating near the apex (Figure 9); at 60° the flow deflects gradually and the stagnation footprint is smaller. The wake follows in step (Figure 8): the wide separation region behind the 150° cone suppresses pressure recovery and holds the base pressure low, while the narrower wake of the 60° cone recovers more. High  $C_d$  at large  $\alpha$  is the product of maximum compression at the front and maximum suction at the base, and the two diminish together as the cone sharpens [3].

This is what Newtonian impact theory discards. Its shadow hypothesis treats the region behind the body as inert, with no dynamics of its own. The computed fields show instead an active recirculation zone drawing fluid back toward the base, and a stagnation field on the inflow side in which the flow deflects before reaching the surface at all. Drag is therefore not a function of the local inflow geometry alone; it also depends on how wide a wake the body forms, which is precisely the mechanism Equation (7) omits. That is the reason for the base term  $b$  in Equation (8).

## 6.4 Semi-empirical model

Writing  $x(\alpha) = \sin^2(\alpha/2)$  makes Equation (8) linear in  $x$ . The experimental uncertainties are heteroscedastic (Table 1), so the experimental fit uses weighted least squares (WLS) with  $w_i = 1/\sigma_i^2$ , where  $\sigma_i$  is the absolute uncertainty  $\delta C_d$  of point  $i$  [13]. The objective function and the weighted coefficient of determination are given in Appendix B. The optimum is

$$a^* = 0.6956, \quad b^* = 0.4796, \quad R_w^2 = 0.98879,$$

so the linear form in  $x$  tracks the data closely over the range tested. The positive intercept means that drag does not vanish as  $x \rightarrow 0$ , which is the qualitative behaviour Section 2.3 anticipated and which Equation (7) cannot produce. A scan of the  $(a, b)$  plane, mapping  $R_w^2$  over a  $500 \times 500$  grid, is shown in Figure 11.

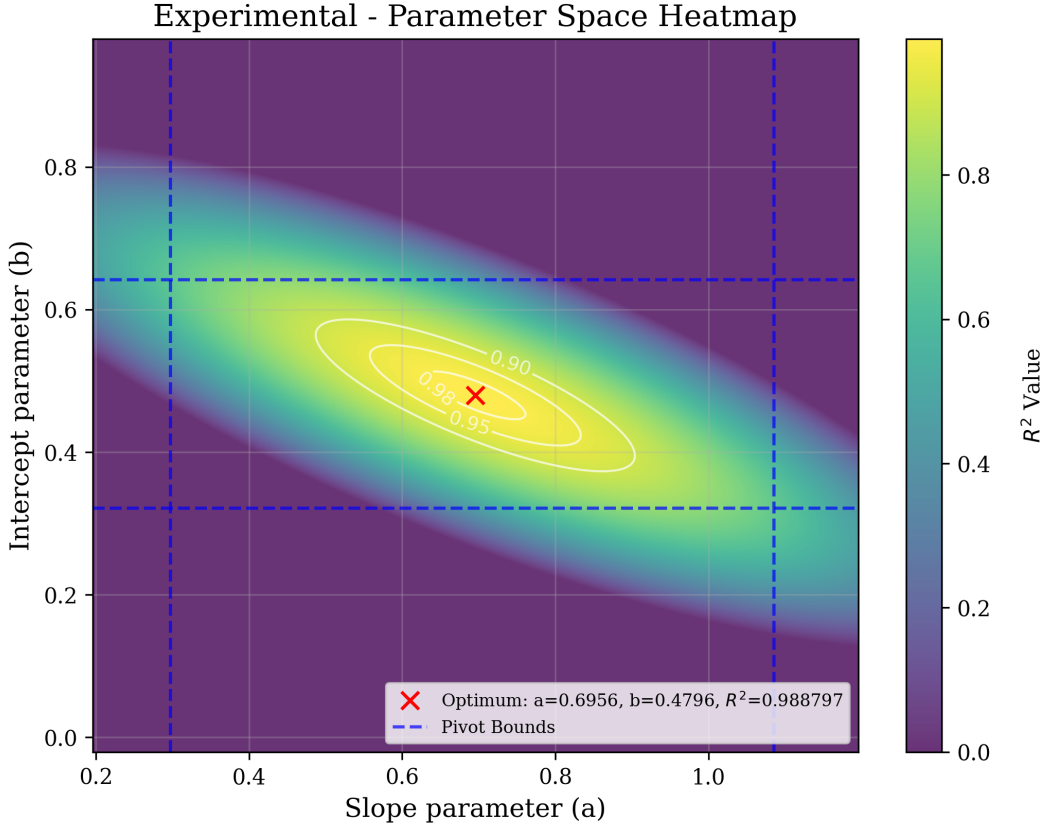


Figure 11: Weighted coefficient of determination over the  $(a, b)$  parameter plane for the experimental fit, with contours at  $R^2 = 0.9, 0.95$  and  $0.98$  and the optimum  $(a^*, b^*)$ .

With only  $N = 5$  points, confidence intervals from the Student  $t$ -distribution are too wide to carry physical meaning, so the parameter envelope was instead obtained from the End-Points Pivot Rule [13], derived in Appendix B. The rule rotates the regression line about the weighted centroid  $(\bar{x}_w, \bar{y}_w)$  and retains the steepest and shallowest slopes still compatible with the error bars on the leftmost and rightmost points. This gives

$$a_{\min} = 0.3450, \quad a_{\max} = 1.0859.$$

The intercept is not free once the slope is set: the line must still pass through the centroid, so

$$\bar{y}_w = a_{\text{new}}\bar{x}_w + b_{\text{new}} \quad \Rightarrow \quad b_{\text{new}} = \bar{y}_w - a_{\text{new}}\bar{x}_w. \quad (12)$$

With  $\bar{x}_w = 0.4062$  and  $\bar{y}_w = 0.7621$ , Equation (12) gives  $b(a_{\min}) = 0.6220$  and  $b(a_{\max}) = 0.3210$ . The experimental model is therefore

$$C_{d,\text{exp}}(\alpha) = 0.6956 \sin^2\left(\frac{\alpha}{2}\right) + 0.4796, \quad (13)$$

with  $a \in [0.3450, 1.0859]$  and  $b \in [0.3210, 0.6220]$  (Figure 12).

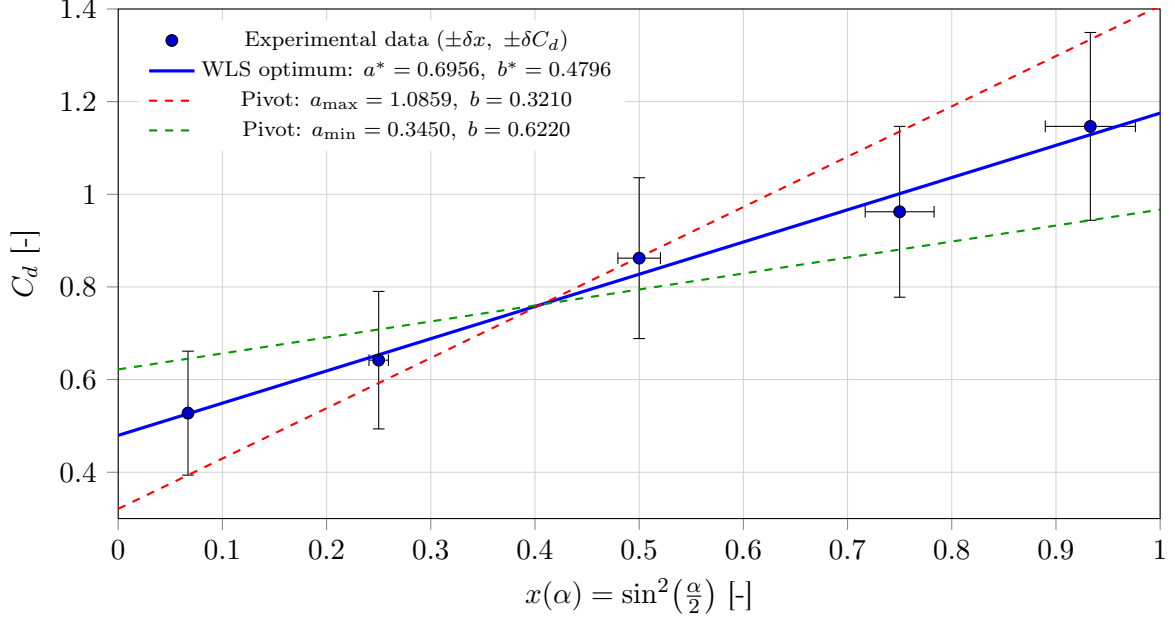


Figure 12: Experimental drag coefficient against  $x(\alpha) = \sin^2(\alpha/2)$  with the weighted least squares fit and the End-Points Pivot Rule envelope.

The computed dataset was fitted with ordinary least squares (OLS,  $w_i = 1$ ), because the CFD points carry no statistical uncertainty and no heteroscedastic  $\sigma_i$  exists to weight them. The fit is descriptive, a concise summary of the numerical trend rather than an estimate of parameter uncertainty [13]. It gives

$$a^* = 0.8775, \quad b^* = 0.3791, \quad R^2 = 0.98139,$$

that is

$$C_{d,\text{CFD}}(\alpha) = 0.8775 \sin^2\left(\frac{\alpha}{2}\right) + 0.3791. \quad (14)$$

The computed slope exceeds the measured one, so the simulation predicts a faster rise of  $C_d$  with bluntness, while its lower intercept indicates less base and skin drag in the limit  $x \rightarrow 0$  for an idealised geometry. The comparison that matters is that

$$a_{\text{CFD}} = 0.8775 \in [0.3450, 1.0859],$$

so the numerical slope lies inside the band the experimental uncertainties allow, and there is no contradiction between the two trends (Figure 13).

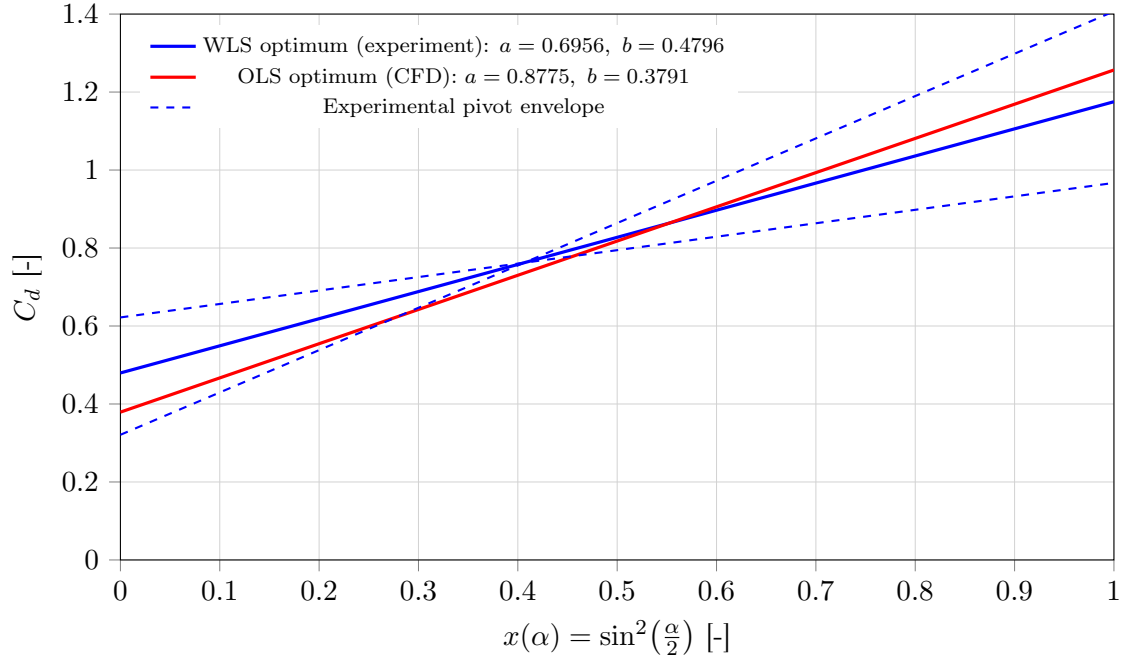


Figure 13: Experimental and computed semi-empirical fits in the  $x(\alpha) = \sin^2(\alpha/2)$  plane, with the experimental pivot envelope.

### 6.5 Verification of the velocity scaling

Equation (2) predicts  $v_t = Kx$  with  $x \equiv C_d^{-1/2}$  and  $K = \sqrt{2mg/(\rho A)} = 2.6952$  for the present constants. Propagating the drag uncertainty through the power transformation gives  $\delta x/x = \frac{1}{2} \delta C_d/C_d$ , by logarithmic differentiation [12]. Figure 14 plots the measured  $v_t$  against  $x$  with uncertainties on both axes; the data lie on the predicted line through the origin across the whole range. The quadratic drag regime therefore holds throughout, and viscous scale effects are negligible for these kinematics [1], [5, pp. 75–89], [3].

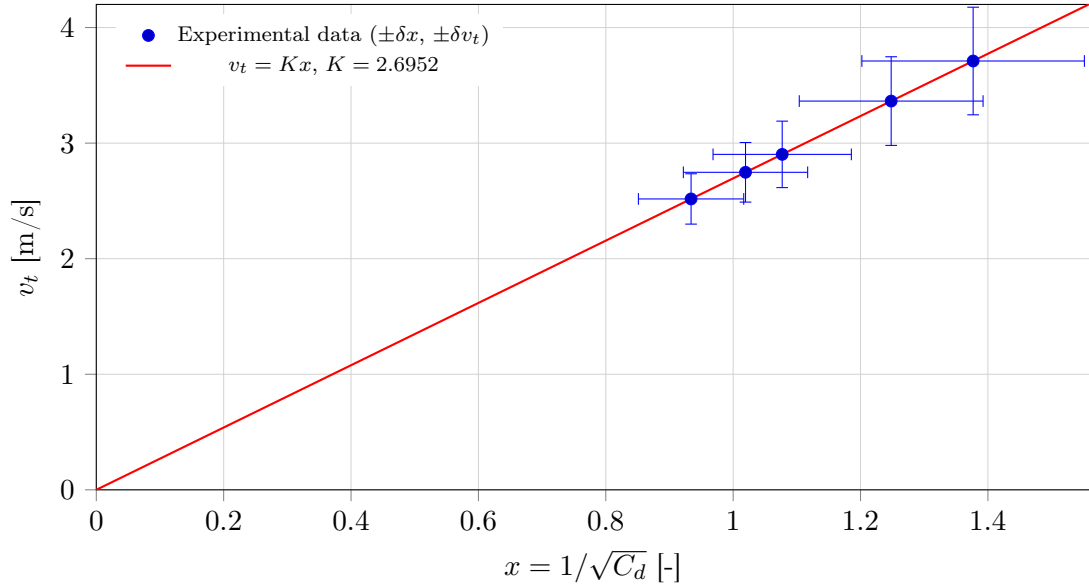


Figure 14: Measured terminal velocity against  $C_d^{-1/2}$  with the predicted line through the origin.

## 7 Conclusions

The internal apex angle governs the drag of a hollow cone falling apex-down. Over the range tested, terminal velocity falls from 3.71 m/s at 30° to 2.52 m/s at 150° while  $C_d$  rises from 0.528 to 1.147, with mass and frontal area held constant. The mechanism is form drag: blunt shells hold near-stagnation pressure across the whole windward face and generate a wide recirculation region that suppresses base pressure recovery, and both effects weaken together as the cone sharpens. Across the full range the measured terminal velocity follows  $v_t \propto C_d^{-1/2}$ , which confirms that the flow stays in the quadratic drag regime and that  $C_d$  is set by the geometrically fixed separation rather than by Reynolds number.

Newtonian impact theory,  $C_d = 4 \sin^2(\alpha/2)$ , is insufficient here because it assigns no pressure to the base and no dynamics to the wake. The corrected two-parameter form fits the measurements with  $R^2 \approx 0.99$ :

$$C_{d,\text{exp}}(\alpha) = 0.6956 \sin^2\left(\frac{\alpha}{2}\right) + 0.4796, \quad \alpha \in [0.3450, 1.0859].$$

The strictly positive intercept is the quantitative result that Newtonian theory cannot reproduce: a baseline of viscous and base-pressure drag that survives as the cone becomes slender.

Steady RANS with the  $k\text{-}\omega$  SST closure reproduced the measurements to  $\lesssim 2\%$  for three of the five geometries and to 8.1% at 120°, all inside the experimental uncertainty, and its fitted trend  $C_{d,\text{CFD}}(\alpha) = 0.8775 \sin^2(\alpha/2) + 0.3791$  has a slope lying within the envelope the measurements allow. The exception is the 30° cone, where the computed  $C_d \approx 0.38$  falls 27% below the measured  $C_d \approx 0.53$ . That gap is attributed to fluid-structure interaction: the 30° paper shell is the least rigid of the set by construction, and its micro-vibration in free fall widens the time-averaged wake in a way that a steady, rigid-geometry solution has no mechanism to represent.

### Limitations and further work

The results hold for thin-walled hollow shells falling apex-down in approximately incompressible subsonic flow, using only stable or quasi-steady trials, and the numerical work does not model unsteadiness or flow-structure interaction. Four steps follow directly:

1. Fabricate rigid specimens, for example 3D-printed shells, to remove aeroelastic flutter. This isolates the pure aerodynamic drag and is the direct test of the 30° interpretation given in Section 6.2.
2. Record at a higher frame rate, which reduces the dominant instrumental uncertainty  $\delta v$  and narrows the confidence intervals on every measured  $C_d$ .
3. Add physical flow visualisation, by particle image velocimetry (PIV) or wind-tunnel smoke tracing, to check the wake topologies and separation locations predicted by the RANS solutions.
4. Extend the computation to unsteady RANS (URANS) to resolve the transient vortex shedding and wake oscillation that the steady formulation averages away.

### Data availability

The raw kinematic data, the CFD cases and post-processing output, and the Python analysis scripts are available at <https://github.com/KRGulaj/conical-shells-repo>.

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## A Raw experimental data log

This appendix lists all 10 trials recorded for each cone geometry. The acceptance protocol of Section 3.3 was applied: the first six trials showing stable descent were used to compute the mean terminal velocity  $v_t$ , and the remainder were censored once that quota was met. Where a specimen diverged transversely before reaching the lowest fiducial marker, the transit time for the B–C interval could not be resolved optically and is denoted by a dash. Rows in bold are the accepted trials.

Internal apex angle: 30°

| Trial | Frames (A-B / B-C) | $v_{mean}$ [m/s] | Status                           |
|-------|--------------------|------------------|----------------------------------|
| 1     | 8 / 8              | 3.75             | Included                         |
| 2     | 8 / 8              | 3.75             | Included                         |
| 3     | 8 / 8              | 3.75             | Included                         |
| 4     | 9 / 8              | 3.53             | Included                         |
| 5     | 8 / -              | -                | Excluded (trajectory divergence) |
| 6     | 8 / 8              | 3.75             | Included                         |
| 7     | 8 / 8              | 3.75             | Included                         |
| 8     | 9 / 8              | 3.53             | Censored (sample quota met)      |
| 9     | 8 / 9              | 3.53             | Censored (sample quota met)      |
| 10    | 8 / 8              | 3.75             | Censored (sample quota met)      |

Internal apex angle: 60°

| Trial | Frames (A-B / B-C) | $v_{mean}$ [m/s] | Status                      |
|-------|--------------------|------------------|-----------------------------|
| 1     | 9 / 9              | 3.33             | Included                    |
| 2     | 9 / 8              | 3.53             | Included                    |
| 3     | 9 / 9              | 3.33             | Included                    |
| 4     | 9 / 9              | 3.33             | Included                    |
| 5     | 9 / 9              | 3.33             | Included                    |
| 6     | 9 / 9              | 3.33             | Included                    |
| 7     | 9 / 8              | 3.53             | Censored (sample quota met) |
| 8     | 8 / -              | -                | Censored (sample quota met) |
| 9     | 9 / 9              | 3.33             | Censored (sample quota met) |
| 10    | 9 / 8              | 3.53             | Censored (sample quota met) |

Internal apex angle: 90°

| Trial | Frames (A-B / B-C) | $v_{mean}$ [m/s] | Status                           |
|-------|--------------------|------------------|----------------------------------|
| 1     | 10 / 10            | 3.00             | Included                         |
| 2     | 10 / 10            | 3.00             | Included                         |
| 3     | 11 / 10            | 2.86             | Included                         |
| 4     | 10 / -             | -                | Excluded (trajectory divergence) |
| 5     | 10 / 11            | 2.86             | Included                         |
| 6     | 11 / 11            | 2.73             | Included                         |
| 7     | 10 / 10            | 3.00             | Included                         |
| 8     | 10 / 11            | 2.86             | Censored (sample quota met)      |
| 9     | 10 / 10            | 3.00             | Censored (sample quota met)      |
| 10    | 11 / 10            | 2.86             | Censored (sample quota met)      |

Internal apex angle: 120°

| Trial | Frames (A-B / B-C) | $v_{mean}$ [m/s] | Status                           |
|-------|--------------------|------------------|----------------------------------|
| 1     | 11 / 11            | 2.73             | Included                         |
| 2     | 11 / 11            | 2.73             | Included                         |
| 3     | 10 / 11            | 2.86             | Included                         |
| 4     | 11 / 11            | 2.73             | Included                         |
| 5     | 11 / 12            | 2.61             | Excluded (kinematic instability) |
| 6     | 10 / 11            | 2.86             | Included                         |
| 7     | 12 / 11            | 2.61             | Included                         |
| 8     | 11 / 10            | 2.86             | Censored (sample quota met)      |
| 9     | 11 / 11            | 2.73             | Censored (sample quota met)      |
| 10    | 12 / 11            | 2.61             | Censored (sample quota met)      |

Internal apex angle: 150°

| Trial | Frames (A-B / B-C) | $v_{mean}$ [m/s] | Status                           |
|-------|--------------------|------------------|----------------------------------|
| 1     | 12 / -             | -                | Excluded (trajectory divergence) |
| 2     | 13 / 12            | 2.40             | Included                         |
| 3     | 12 / 12            | 2.50             | Included                         |
| 4     | 12 / 13            | 2.40             | Excluded (kinematic instability) |
| 5     | 12 / 12            | 2.50             | Included                         |
| 6     | 12 / 12            | 2.50             | Included                         |
| 7     | 12 / 12            | 2.50             | Included                         |
| 8     | 11 / 11            | 2.73             | Included                         |
| 9     | 11 / 12            | 2.61             | Censored (sample quota met)      |
| 10    | 12 / 13            | 2.40             | Censored (sample quota met)      |

## B Regression method and parameter-space scan

### Overview

The semi-empirical parameters  $a$  and  $b$  of the drag model  $C_d(\alpha) = a \sin^2(\alpha/2) + b$  were obtained by two regression techniques, chosen according to the nature of the data [13]:

1. weighted least squares (WLS) for the experimental data, which accounts for the heteroscedastic measurement uncertainty  $\sigma_i$ ;
2. ordinary least squares (OLS) for the CFD data, for which no statistical uncertainty is quantified.

### Weighted least squares (experimental data)

Given the experimental dataset  $\{(x_i, y_i, \sigma_i)\}$ , where  $x_i = \sin^2(\alpha_i/2)$  and  $y_i = C_{d,i}$ , weights were defined as  $w_i = 1/\sigma_i^2$ . The WLS objective function minimizes the weighted sum of squared residuals:

$$S_{wls}(a, b) = \sum_{i=1}^N w_i (y_i - (ax_i + b))^2. \quad (15)$$

The goodness of fit was evaluated using the weighted coefficient of determination:

$$R_w^2 = 1 - \frac{\sum w_i (y_i - \hat{y}_i)^2}{\sum w_i (y_i - \bar{y}_w)^2}, \quad (16)$$

where  $\hat{y}_i = ax_i + b$  are the model predictions and  $\bar{y}_w$  is the weighted mean of the observed data.

### Ordinary least squares (CFD data)

For the CFD  $k\text{-}\omega$  SST results, an unweighted OLS approach was used, equivalent to setting  $w_i = 1$ . The objective function simplifies to:

$$S_{ols}(a, b) = \sum_{i=1}^N (y_i - (ax_i + b))^2. \quad (17)$$

### Parameter bounds: the End-Points Pivot Rule

The bounds on the slope  $a$  for the experimental data were obtained from the End-Points Pivot Rule. The heuristic finds the largest and smallest slopes still admissible by pivoting the regression line about the weighted centroid  $(\bar{x}_w, \bar{y}_w)$  out to the extremes of the error bars on the first (leftmost) and last (rightmost) data points.

The bounding slopes follow from the extreme limits of those two terminal points:

$$a_{bound} = \frac{(y_{end} \pm \sigma_{end}) - \bar{y}_w}{x_{end} - \bar{x}_w}. \quad (18)$$

The minimum and maximum of these slopes define the admissible range  $[a_{min}, a_{max}]$ , which measures how far the fitted line can be tilted by the most constraining measurement uncertainties.

### Numerical implementation

The Python routine written for this analysis executes the following sequence:

1. defines the parameter-space grid at  $500 \times 500$  resolution;
2. computes the  $R^2$  scalar field for both the WLS (experimental) and OLS (CFD) cases;

3. extracts the optimal parameters  $(a, b)$  at the maximum  $R^2$ ;
4. calculates the experimental slope bounds from the End-Points Pivot Rule;
5. exports the results to .dat files for plotting.

The slope bounds  $[a_{min}, a_{max}]$  returned by the script are physically meaningful only for the experimental data, where  $\sigma_i$  is a measured uncertainty. For the CFD dataset, where no statistical uncertainty is quantified and  $\sigma_i \approx 0$ , the pivot rule collapses onto the single OLS solution, and the script is configured to skip the bounds output for that case.

```

1 import numpy as np
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 from pathlib import Path
5
6 # Raw Data Constants
7 X_DATA = np.array([0.06698729810778929, 0.25, 0.5, 0.75, 0.9330127018922625])
8 EXP_Y_DATA = np.array([0.527576297466166, 0.6419620607599249, 0.862154655100411,
9     0.9622421979824501, 1.146605134114744])
10 EXP_SIGMA = np.array([0.13379016000002, 0.14838547909538, 0.17356698767590,
11     0.18439605617689, 0.20269902398342])
12 CFD_Y_DATA = np.array([0.383698983, 0.64161545, 0.857249858, 1.039745268, 1.16709476])
13
14 # Parameter Space Generator
15 def generate_parameter_space(a_opt, b_opt, a_span=0.5, b_span=0.5, resolution=500):
16     """Generate dynamic parameter grid for slope (a) and intercept (b)."""
17     a_min = a_opt - a_span
18     a_max = a_opt + a_span
19     b_min = b_opt - b_span
20     b_max = b_opt + b_span
21
22     a_values = np.linspace(a_min, a_max, resolution)
23     b_values = np.linspace(b_min, b_max, resolution)
24     A_GRID, B_GRID = np.meshgrid(a_values, b_values)
25     return A_GRID, B_GRID
26
27 # Analytical Solvers
28 def analytical_wls(x, y, weights):
29     """
30     Calculate exact WLS parameters using closed-form formulas.
31     Returns optimal a, b, and the true maximum R^2 value.
32     """
33     w_sum = np.sum(weights)
34
35     # Weighted means
36     x_bar = np.sum(weights * x) / w_sum
37     y_bar = np.sum(weights * y) / w_sum
38
39     # Calculate slope (a) using wls
40     numerator = np.sum(weights * (x - x_bar) * (y - y_bar))
41     denominator = np.sum(weights * (x - x_bar)**2)
42
43     a_opt = numerator / denominator
44     b_opt = y_bar - a_opt * x_bar
45
46     # Calculate exact true maximum R^2
47     ss_tot_weighted = np.sum(weights * (y - y_bar)**2)
48     y_pred = a_opt * x + b_opt
49     ss_res_weighted = np.sum(weights * (y - y_pred)**2)
50
51     r2_max = 1.0 - (ss_res_weighted / ss_tot_weighted) if ss_tot_weighted > 0 else 0.0
52
53     return a_opt, b_opt, r2_max
54
55 # Analytical Solvers
56 def pivot_rule_bounds(x, y, sigma, weights):
57     """
58     Calculate theoretical slope bounds based on error bars bounding the centroid.
59     Uses only first (leftmost) and last (rightmost) points.
60     Returns a_min, a_max, b_min, b_max for the smallest valid interval.
61     """

```

```

60     w_sum = np.sum(weights)
61
62     # Weighted centroid
63     x_centroid = np.sum(weights * x) / w_sum
64     y_centroid = np.sum(weights * y) / w_sum
65
66     # First point (leftmost) and last point (rightmost)
67     x_left, y_left, sigma_left = x[0], y[0], sigma[0]
68     x_right, y_right, sigma_right = x[-1], y[-1], sigma[-1]
69
70     # Calculate dx from centroid to each extreme point
71     dx_L = x_left - x_centroid
72     dx_R = x_right - x_centroid
73
74     # Calculate slopes from centroid to error boundaries of left point
75     slope_L_min = (y_left - sigma_left - y_centroid) / dx_L if dx_L != 0 else 0.0
76     slope_L_max = (y_left + sigma_left - y_centroid) / dx_L if dx_L != 0 else 0.0
77
78     # Calculate slopes from centroid to error boundaries of right point
79     slope_R_min = (y_right - sigma_right - y_centroid) / dx_R if dx_R != 0 else 0.0
80     slope_R_max = (y_right + sigma_right - y_centroid) / dx_R if dx_R != 0 else 0.0
81
82     # Return the smallest interval across all bounds
83     a_min = max(slope_L_min, slope_R_min)
84     a_max = min(slope_L_max, slope_R_max)
85
86     # Calculate corresponding intercepts
87     b_min = y_centroid - a_max * x_centroid
88     b_max = y_centroid - a_min * x_centroid
89
90     return a_min, a_max, b_min, b_max
91
92
93 # R^2 Grid Calculator
94 def compute_r2_surface(x, y, weights, a_grid, b_grid):
95     """Calculate R^2 surface across the entire parameter grid."""
96     w_sum = np.sum(weights)
97     y_bar = np.sum(weights * y) / w_sum
98
99     # Total Sum of Squares (weighted)
100     ss_tot = np.sum(weights * (y - y_bar)**2)
101
102     # Calculate predictions for entire grid using broadcasting
103     preds = a_grid[:, :, None] * x[None, None, :] + b_grid[:, :, None]
104     # Residual Sum of Squares (weighted)
105     ss_res = np.sum(weights[None, None, :] * (y[None, None, :] - preds)**2, axis=2)
106
107     # Calculate R^2
108     r2_array = 1.0 - (ss_res / ss_tot)
109
110     # Clip the negative values
111     r2_array = np.clip(r2_array, a_min=0, a_max=1)
112
113     return r2_array
114
115 # Visualization
116
117 def analyze_and_plot(a_grid, b_grid, r2_surface, a_opt, b_opt, r2_opt,
118                     pivot_bounds=None, data_label="Data", output_dir="."):
119
120     plt.rcParams['font.family'] = 'serif'
121     fig, ax = plt.subplots(figsize=(8, 6))
122
123     # Dynamically extract extent from grid min/max values
124     extent = [a_grid.min(), a_grid.max(), b_grid.min(), b_grid.max()]
125
126     # Plot heatmap
127     im = ax.imshow(r2_surface, extent=extent, origin='lower', aspect='auto',
128                   cmap='viridis', alpha=0.8)
129
130     # R^2 contours
131     contour_levels = [0.90, 0.95, 0.98]
132     cs = ax.contour(a_grid, b_grid, r2_surface, levels=contour_levels,

```

```

133         colors='white', linewidths=1, alpha=0.7)
134 ax.clabel(cs, inline=True, fontsize=9, fmt=r'${$.2f}$')
135
136 # Analytical optimum
137 ax.plot(a_opt, b_opt, 'rx', markersize=8, markeredgewidth=1.5,
138         label=f'Optimum: a={a_opt:.4f}, b={b_opt:.4f}, $R^2$={r2_opt:.6f}')
139
140
141 # Add pivot rule bounds if available
142 if pivot_bounds is not None:
143     a_min_bound, a_max_bound = pivot_bounds[0], pivot_bounds[1]
144
145     # Vertical lines for slope bounds (a)
146     ax.axvline(x=a_min_bound, color='blue', linestyle='--', linewidth=1.5,
147               alpha=0.7)#
148     ax.axvline(x=a_max_bound, color='blue', linestyle='--', linewidth=1.5,
149               alpha=0.7, label='Pivot Bounds')
150
151     # Horizontal lines for intercept bounds (b)
152     if len(pivot_bounds) >= 4:
153         b_min_bound, b_max_bound = pivot_bounds[2], pivot_bounds[3]
154         ax.axhline(y=b_min_bound, color='blue', linestyle='--', linewidth=1.5,
155                   alpha=0.7)
156         ax.axhline(y=b_max_bound, color='blue', linestyle='--', linewidth=1.5,
157                   alpha=0.7)
158
159 # Labels and title
160 ax.set_xlabel('Slope parameter (a)', fontsize=12)
161 ax.set_ylabel('Intercept parameter (b)', fontsize=12)
162 ax.set_title(f'{data_label} - Parameter Space Heatmap',
163             fontsize=14)
164
165 # Colorbar
166 cbar = plt.colorbar(im, ax=ax)
167 cbar.set_label('$R^2$ Value', rotation=90, labelpad=15)
168
169 # Legend
170 ax.legend(loc='lower right', fontsize=8)
171
172 # Grid
173 ax.grid(True, alpha=0.3)
174
175 # Save figure
176 output_path = Path(output_dir) / f'{data_label}_parameter_space_heatmap.png'
177 plt.savefig(output_path, dpi=300, bbox_inches='tight', facecolor='white')
178 print(f"Saved heatmap: {output_path}")
179 plt.close(fig)
180
181 # Flatten grids and save to .dat CSV file
182 a_flat = a_grid.flatten()
183 b_flat = b_grid.flatten()
184 r2_flat = r2_surface.flatten()
185
186 data_dict = {'a': a_flat, 'b': b_flat, 'R2': r2_flat}
187
188 # execution directory
189 csv_path = Path(output_dir) / f'{data_label}_parameter_space.dat'
190
191 df = pd.DataFrame({'a': a_flat, 'b': b_flat, 'R2': r2_flat})
192 df.to_csv(csv_path, index=False, float_format='%.6f')
193
194 print(f"Saved CSV data: {csv_path}")
195
196
197
198 # Execute Pipelines
199 if __name__ == "__main__":
200     # Get script execution directory
201     script_dir = Path(__file__).parent
202
203     # Experimental Data (WLS with 1/sigma^2 weights)
204     print("EXPERIMENTAL DATA ANALYSIS")
205

```

```

206     exp_weights = 1.0 / (EXP_SIGMA ** 2)
207
208     # Run WLS analytical solver
209     a_exp_opt, b_exp_opt, r2_exp_opt = analytical_wls(X_DATA, EXP_Y_DATA, exp_weights)
210     print(f"Analytical Optimum: a={a_exp_opt:.6f}, b={b_exp_opt:.6f}")
211     print(f"Maximum  $R^2$ : {r2_exp_opt:.6f}")
212
213     # Generate dynamic grid for experimental data
214     A_GRID_EXP, B_GRID_EXP = generate_parameter_space(a_exp_opt, b_exp_opt)
215
216     # Run  $R^2$  grid calculator
217     r2_exp_surface = compute_r2_surface(X_DATA, EXP_Y_DATA, exp_weights, A_GRID_EXP,
218                                         B_GRID_EXP)
219
220     # Extract pivot bounds
221     pivot_bounds_exp = pivot_rule_bounds(X_DATA, EXP_Y_DATA, EXP_SIGMA, exp_weights)
222     print(f"Pivot Rule Bounds:")
223     print(f"a_min={pivot_bounds_exp[0]:.6f}, b_min={pivot_bounds_exp[2]:.6f}")
224     print(f"a_max={pivot_bounds_exp[1]:.6f}, b_max={pivot_bounds_exp[3]:.6f}")
225
226     # Send to plotter
227     analyze_and_plot(A_GRID_EXP, B_GRID_EXP, r2_exp_surface, a_exp_opt, b_exp_opt,
228                     r2_exp_opt, pivot_bounds=pivot_bounds_exp,
229                     data_label="Experimental", output_dir=script_dir)
230
231     # CFD Data (OLS with weights = 1.0)
232     print("CFD DATA ANALYSIS")
233
234     cfd_weights = np.ones_like(CFD_Y_DATA)
235     cfd_sigma = np.zeros_like(CFD_Y_DATA)
236
237     # Run analytical solver (OLS equivalent with uniform weights)
238     a_cfd_opt, b_cfd_opt, r2_cfd_opt = analytical_wls(X_DATA, CFD_Y_DATA, cfd_weights)
239     print(f"Analytical Optimum: a={a_cfd_opt:.6f}, b={b_cfd_opt:.6f}")
240     print(f"Maximum  $R^2$ : {r2_cfd_opt:.6f}")
241
242     # Generate dynamic grid for CFD data
243     A_GRID_CFD, B_GRID_CFD = generate_parameter_space(a_cfd_opt, b_cfd_opt)
244
245     # Run  $R^2$  grid calculator
246     r2_cfd_surface = compute_r2_surface(X_DATA, CFD_Y_DATA, cfd_weights, A_GRID_CFD,
247                                         B_GRID_CFD)
248
249     # Send to plotter (skipping pivot bounds for CFD)
250     analyze_and_plot(A_GRID_CFD, B_GRID_CFD, r2_cfd_surface, a_cfd_opt, b_cfd_opt,
251                     r2_cfd_opt, pivot_bounds=None, data_label="CFD",
252                     output_dir=script_dir)
253
254     print("ANALYSIS COMPLETE")

```

Listing 1: Python script used to compute WLS/OLS optima, Pivot Rule bounds, and generate parameter-space heatmaps.